Definitions and properties.

1. Directional derivative : $f'(x,v) = \lim_{t\to 0} \frac{f(x+tv)-f(x)}{t} = \nabla f(x)^T v$

2. Stationary point $x^* \to \nabla f(x^*) = 0$.

① x^* local min $\to \nabla f(x^*) = 0$, $\nabla^2 f(x) \succeq 0$

② $\nabla f(x^*) \succ 0$, $\nabla f(x^*) = 0 \to x^*$ local min

3. Descent direction $p \to \nabla f(x)^T p < 0$, steepest $\to p = -\frac{\nabla f(x)}{\|\nabla f(x)\|}$

4. L-Lipschitz : $\| f(u) - f(v) \| \le L \cdot \|u - v\|$; L-smooth : ∇f L-Li~

f is L-S··· : ① $f(v) \le f(u) + g(u)^T (v-u) + \frac{L}{2}\|v-u\|^2$

descent Lemma $f(v) \ge f(u) + g(u)^T (v-u) - \frac{L}{2}\|v-u\|^2$

② $x_{k+1} = x_k - \alpha g_k$, $\alpha < \frac{2}{L}$, $f(x_k) - f(x_{k+1}) \ge \alpha(1 - \frac{\alpha L}{2})\|g_k\|^2$

③ $f^* = \inf f > -\infty$, $\alpha = \frac{1}{L}$, $\min_i \|g_i\|^2 \le \frac{2L}{k}(f(x_0) - f^*) = O(\frac{1}{k})$

5. (strict) convex : $\forall \theta \in [0,1]$, $f(u + (1-\theta)v)$ $[<] \le \theta f(u) + (1-\theta) f(v)$

f is convex : ① $H_S \succeq 0$ ($H_S \succ 0$ when strict convex)

② $f(v) - f(u) \ge g(u)^T (v-u) \leftrightarrow (g(v) - g(u))^T (v-u) \ge 0 \leftrightarrow$ convex

③ $\nabla f(x^*) = 0 \leftrightarrow x^*$ is local min

④ L-smooth, $\alpha = \frac{1}{L}$, $\exists x^*$: $f(x_k) - f(x^*) \le \frac{L}{2k} \cdot \|x_0 - x^*\|^2 = O(\frac{1}{k})$

6. μ-strong convex : $f(v) - f(u) \ge g(u)^T (v-u) + \frac{\mu}{2}\|v-u\|^2$

Condiction number $K = \frac{L}{\mu} \ge 1$ ① μ/L : $\min/\max$ eigenvale of H_S

② $L, \mu \leftrightarrow \mu I \preceq \nabla^2 f(x) \preceq LI$ ③ μ : $f(x) - f^* \le \frac{1}{2\mu}\|g\|^2$

④ μ, x^* unique : $\|g\| \ge \mu \|x - x^*\|$

⑤ $\mu, L, \alpha = \frac{1}{L}$: $f(x_k) - f^* \le (1 - \frac{\mu}{L})^k (f(x_0) - f^*) = O((1-K)^k)$

7.

Theorems and Algorithms

1. **Weierstrass Theorem**: f continuous, compact, non-empty $\to$ local min exist.
2. **Chain Rule**: $F=g\circ h$, $\nabla F(x)=J_h(x)^T\nabla g(h(x))$, $J_h(x)=(\nabla h_1(x)^T,\dots)$
3. **Fundamental Theorem of Calculus**: $f(v)-f(u)=\int_0^1 \nabla f(u+t(v-u))^T(v-u)\,dt$
4. **Taylor Theorem**: $f(x+s)=f(x)+\nabla f(x)^Ts+o(s)=f(x)+\nabla f(x)^Ts+\frac{1}{2}s^TH_xs+o(s^2)$
5. **Gradient Descent**: $x_{k+1}=x_k-\alpha_k\nabla f(x_k)$
6. **Newton Step**: $m_k^{(2)}(p)=f(x_k)+g_k^Tp+\frac{1}{2}p^TH_kp$.

   Let $\nabla_p m=0\to H_kp_k=-g_k$, $\alpha_{k+1}=x_k+p_k\to$ Newton direction.

① $H_{x^*}\succ 0$, H_x ρ-lipschitz near x^*, $x_0\to x^*$: $\exists C$, $\|x_{k+1}-x^*\|\le C\|x_k-x^*\|$

② if ① $H_{x^*}\succ 0$, ② f is quadratic $\to$ NS minimize by 1 step.

7. **Line Search**: chose $g_k^Tp_k<0$, $x_{k+1}=x_k+\alpha_kp_k$.

   Armijo Condition: fix $c_A\in(0,1)$, usually 10^{-4}, $\alpha_k>0$ acceptable iff

   $f(x_k)-f(x_{k+1})\ge -c_A\alpha g_k^Tp_k$. usually $\bar\alpha=1$, $\eta=\frac{1}{2}$

   Backtracking LS: from $\alpha_k^{(0)}>0$, $\alpha_k^{(t+1)}\leftarrow\eta\alpha_k^{(t)}$ until armijo.

① **Exact line search on quadratic**: $f(x)=\frac{1}{2}x^TQx+b^Tx\to x^*=\frac{g^Tg}{g^TQg}$

② $\exists\alpha_0>0$, $\forall\alpha\in(0,\alpha_0]$ armijo ② **global convergence with BLS**:

$\exists\inf f$, L, $g_k^Tp_k\le-\sigma\|g_k\|^2$, $\|p_k\|\le\beta\|g\|$, c_A $\bar\alpha$ of armijo:

$\alpha=\min\{\bar\alpha,2\eta(1-c_A)\sigma\cdot\frac{1}{L\beta^2}\}$, $\min_j\|g_j\|^2\le\frac{f(x_0)-f^*}{c_A\sigma\alpha\cdot k}$, $\|g_k\|\to 0$

④ μ, L, $p_k=-H_k^{-1}g_k$: $g_k^Tp_k\le-\frac{1}{L}\|g_k\|^2$, $\|p_k\|\le\frac{1}{\mu}\|g_k\|$ $(\sigma=\frac{1}{L},\beta=\frac{1}{\mu})$

⑤ L,μ, Newton direction and BLS: $x_k\to 0$ from $\forall x_0$

⑥ If $\bar\alpha=1$, $c_A<\frac{1}{2}$, $H_k(x^*)$ local ρ-Lip: $\alpha_k=1$ finally